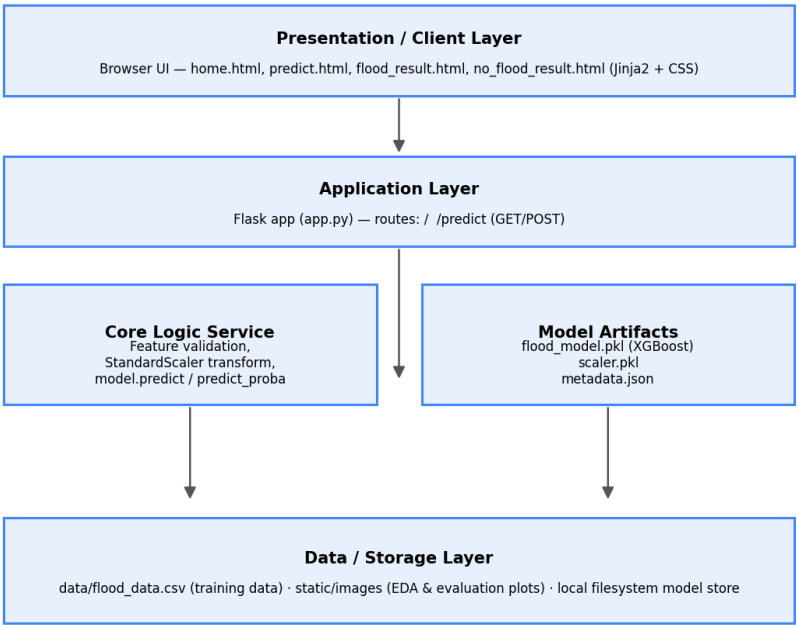


Solution Architecture

Date	1 July 2026
Team ID	—
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

The diagram below maps Rising Waters' Presentation, Application and Data layers to the specific technologies used in the project.

Solution Architecture — Rising Waters Flood Prediction System



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the home, prediction form and result pages that residents/officials interact with	HTML5, Jinja2 templates, CSS
Application Layer	Flask app handling routing, form submission and orchestrating the prediction request	Python, Flask
Core Logic Service	Validates inputs, applies the saved StandardScaler, and runs the trained classifier	scikit-learn, XGBoost, pandas, numpy
Model Artifacts / Data	Stores the trained model, scaler and metadata,	joblib (.pkl), JSON, CSV, local

Component Name	Description / Role in Architecture	Technologies Used
Layer	plus the source training dataset and EDA images	filesystem